

Supplementary Material

Design and Analysis of a Robotic Dexterous Hand: Combining Linkage Driven and Pneumatic Actuation

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1 Abbreviations and nomenclature

DoF	degree of freedom
F-E	flexion and extension
A-A	abduction and adduction
P-S	pronation and supination
MCP	Metacarpophalangeal (joint)
PIP	Proximal Interphalangeal (joint)
DIP	Distal Interphalangeal (joint)
TM	Trapeziometacarpal (joint)
IP	Interphalangeal (joint)
$\lambda_i (i = 1, 2, 3)$	principal stretches, i.e., stretch (extension) ratio of a differential cubic volume element along the length, width, and height directions of the chamber, respectively
$I_i (i = 1, 2, 3)$	principal invariants
W	strain energy function
$C_i (i = 1, 2, 3)$	material coefficients (in MPa)
$\omega_i (i = 1, 2, 3)$	length, width, and lateral wall thickness of the chamber (in mm)
h_1	height of the chamber (in mm)
ρ	bending radius of the membrane curvature (in mm)
δ_1	bulge of the membrane under inflation (in mm)
θ_m	clamped angle defined in Fig. 3(b)
λ	stretch of the membrane before contact, $\lambda = \theta_m / \sin\theta_m$
P	air pressure (in MPa)
C	contact initiation point in Fig. 4(b)
d_0	initial distance between the membrane and the motor (in mm)
θ_b	inclination angle of the finger
d	normal distance from the contact plane to the membrane (in mm)
λ^*	stretch of the membrane during contact
c_a, c_b	contact width along the plane in Fig. 4(c) (in mm)
ρ^*	the location of contact point C on the undeformed membrane
ρ_1, ρ_2	initial location of the interior and exterior contact width
ρ_c	radius of the interior and exterior membrane curvature (during contact, in mm)
θ_x, θ_y	interior and exterior clamped angle (during contact)
A_c	transverse contact area between the motor and the membrane
c_z	transverse contact width (in mm)
F_c	contact force generated by the squeezing
M_c	bending moment generated by the contact force
O	rotation origin
e_y	offset from the application point of F_c to the rotation origin (in mm)

F_r	restoring force generated by the preloaded tension spring
M_r	restoring moment generated by F_r
ΔL	variation in length of the spring (in mm)
k	spring elastic coefficient (in N/mm)
$L_i (i = 1, 2, \dots, 11)$	distance between each joint of the hand (in mm)
d_1	distance from the chamber center line to the rotation origin (in mm)

2 Derivations of the moment M_r and the force F_r

The following discussions are given based on Fig. 1(c) and Fig. 4 in the manuscript. Note that all distances between each point are given as projected lengths, not as actual 3D distances. As shown in Fig. 1(c), the coordinates of point G are $(\overline{OG} \cdot \cos \theta_b, \overline{OG} \cdot \sin \theta_b)$. The coordinates of point H are

$$x_H = \overline{OF} - \frac{\overline{FH}}{\overline{FG}} \cdot (\overline{OG} \cdot \cos \theta_b - \overline{OF}) , \quad y_H = -\frac{\overline{FH}}{\overline{FG}} \cdot \overline{OG} \cdot \sin \theta_b \quad (1)$$

where $\overline{OF}$ denotes the length of link OF, with similar notation for the other links.

The length of link FG is variable, and is given as

$$\overline{FG} = \sqrt{(\overline{OG} \cdot \cos \theta_b - \overline{OF})^2 + (\overline{OG} \cdot \sin \theta_b)^2} . \quad (2)$$

The link system as shown in Fig. 1(c) is subjected to the external moment M_c and the force F_r . A virtual rotation $\delta \theta_b$ is imposed on link OG, inducing a virtual displacement $\delta \vec{r}_H = (\delta x_H, \delta y_H)$ at point H. According to the principle of virtual work, we have

$$M_c \cdot \delta \theta_b + \vec{F}_r \cdot \delta \vec{r}_H = 0 \quad (3)$$

where $\vec{F}_r$ is the force vector, and the kinematic relation $\delta \vec{r}_H = (d\vec{r}_H/d\theta_b)\delta \theta_b$ obtained from the geometry is applied to relate the displacements.

Let $\vec{e}_{HE}$ corresponds to the unit vector along link HE, yielding

$$\vec{e}_{HE} = \frac{(\overline{OE} - x_H, -y_H)}{\overline{HE}} . \quad (4)$$

Hence,

$$M_c = -F_r \cdot \frac{\vec{e}_{HE} \cdot d\vec{r}_H}{d\theta_b} . \quad (5)$$

The projected length of links FH, FE, OE, and OF is constant, while OG is obtained

when the F-E motor is at its original position (the exact projected length of OG varies with the finger A-A and F-E motion, and is modeled as a constant for simplicity). The length of HE can be computed using the law of cosines. The length of FG is

$$\overline{FG} = \sqrt{\overline{OG}^2 + \overline{OF}^2 - 2\overline{OG} \cdot \overline{OF} \cdot \cos \theta_b} . \quad (6)$$

The derivative of length FG with respect to the angle θ_b is

$$\frac{d\overline{FG}}{d\theta_b} = \frac{\overline{OG} \cdot \overline{OF} \cdot \sin \theta_b}{\overline{FG}} . \quad (7)$$

The derivative of point H with respect to the angle θ_b is

$$\frac{dx_H}{d\theta_b} = -\frac{\overline{FH}}{\overline{FG}}(-\overline{OG} \cdot \sin \theta_b) + \frac{\overline{FH}(\overline{OG} \cdot \cos \theta_b - \overline{OF})}{\overline{FG}^2} \cdot \frac{d\overline{FG}}{d\theta_b} , \quad (8)$$

$$\frac{dy_H}{d\theta_b} = -\frac{\overline{FH} \cdot \overline{OG} \cdot \cos \theta_b}{\overline{FG}} + \frac{\overline{FH} \cdot \overline{OG} \cdot \sin \theta_b}{\overline{FG}^2} \cdot \frac{d\overline{FG}}{d\theta_b} . \quad (9)$$

Furthermore,

$$\begin{aligned} \overrightarrow{e_{HE}} \cdot \frac{d\overrightarrow{r_H}}{d\theta_b} &= \frac{\overline{OE} - x_H}{\overline{HE}} \cdot \frac{dx_H}{d\theta_b} + \frac{-y_H}{\overline{HE}} \cdot \frac{dy_H}{d\theta_b} \\ &= \frac{\overline{OE} - x_H}{\overline{HE}} \left[\frac{\overline{FH} \cdot \overline{OG} \cdot \sin \theta_b}{\overline{FG}} + \frac{\overline{FH} \cdot \overline{OG} \cdot \overline{OF} \cdot \sin \theta_b (\overline{OG} \cdot \cos \theta_b - \overline{OF})}{\overline{FG}^3} \right] \\ &\quad + \frac{-y_H}{\overline{HE}} \left[-\frac{\overline{FH} \cdot \overline{OG} \cdot \cos \theta_b}{\overline{FG}} + \frac{\overline{FH} \cdot \overline{OG}^2 \cdot \overline{OF} \cdot \sin^2 \theta_b}{\overline{FG}^3} \right] . \end{aligned} \quad (10)$$

Substitute the above equations into the expression of M_c , we have

$$M_c = M_r = -\frac{\overline{FH} \cdot \overline{OG}^2 \cdot (\overline{OE} - \overline{OF}) \cdot \sin \theta_b \cdot (\overline{OG} - \overline{OF} \cdot \cos \theta_b)}{\overline{HE} \cdot \overline{FG}^3} \cdot F_r . \quad (11)$$

Then the relation between the moment M_c and the force F_r is established using the projected length of each link. Detailed derivation on the force F_r is given as follows.

As shown in Fig. 1(c), the S-shaped spring, which is a part of link OG, is modeled as a constant when the finger is not subjected to tip contact. Thus,

$$\angle FGO = \sin^{-1} \left(\frac{\overline{OF} \cdot \sin(\theta_b)}{\overline{FG}} \right) . \quad (12)$$

Based on the geometric relations, we have

$$\angle OFH = \theta_b + \angle FGO , \quad (13)$$

$$\Delta L = \sqrt{\overline{FH}^2 + \overline{FE}^2 - 2\overline{FH} \cdot \overline{FE} \cdot \cos(\angle OFH)} - (\overline{FE} - \overline{FH}) . \quad (14)$$

where length (FE-FH) denotes the spring length before A-A motion.

The restoring force F_r can be calculated though

$$F_r = k \cdot \Delta L \quad (15)$$

where ΔL is the variation in the spring length, and k is the elastic coefficient that ranges from 0.2 N/mm to 0.7 N/mm for illustration (The k value was obtained by pulling the spring with a dynamometer over a measured displacement. The force was read to two decimal places, whereas the displacement was measured with a ruler to one decimal place. Therefore, the raw coefficient $k \approx 0.67$ is rounded to 0.7 to reflect the measurement resolution. Theoretical results of θ_b when k ranges from 0.2 to 0.7 in increments of 0.1 are provided to illustrate the variations across varied inputs. The following table additionally lists the results under 0.8, 0.9, and 1.0 N/mm).

	$P = 0.010$	$P = 0.015$	$P = 0.020$	$P = 0.025$	$P = 0.030$	$P = 0.035$
$k = 0.8$	0.037	0.040	0.042	0.043	0.044	0.045
$k = 0.9$	0.034	0.036	0.037	0.038	0.039	0.040
$k = 1.0$	0.031	0.033	0.034	0.035	0.035	0.036

Hence, the angle θ_b of the finger presented in Table I is computed when the finger is at its stretched state (extension, the F-E motor is at its initial position). The sum of angle θ_b and angle FGO equals the A-A angle of the finger.

3 Definition of the hand matrices

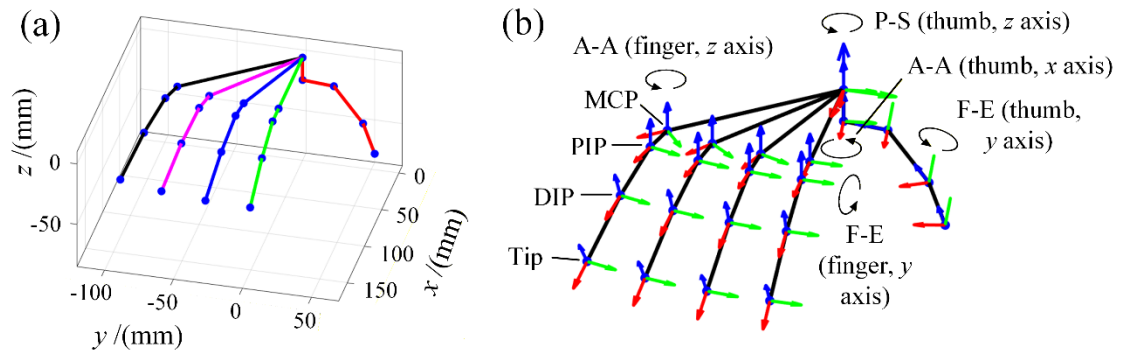


Fig. 1 Schematic of: (a) the hand's initial stretched state; (b) the hand's matrices.

The positive directions of the x , y , and z axes of each frame are shown in red, green, and blue, respectively, in Fig. 1(b) (the one provided in the Supplementary Material). The positive rotation directions are indicated in black.

4 Comparisons among selected 5-fingered dexterous hands

Reference	Year	Actuation strategy	Length, width, and height (in mm)	Active DoF / Joints	External components
Shadow hand [4]	2005*	Tendon	$(265-448) \times (84-135)$	20 / 24	Forearm-required
Pisa/IIT SoftHand 2 [2]	2018	Tendon	Not specified	2 / 19	Forearm-required
ILDA hand [6]	2021	Linkage	$(218-261) \times (102-110) \times (66.8-84.9)$	15 / 20	Wrist-required
BRL/Pisa/IIT SoftHand [3]	2022	Tendon	200×90	1 / 15	Control board
mini X-hand [10]	2022	Tendon, gear, screw and slider	Not specified	2 / 11	Not specified
Ke <i>et al.</i> [18]	2022	Tendon and bevel gear	$185 \times 84 \times (25-42)$	6 / 17	Control board
RBO Hand 3 [24]	2022	Pneumatic actuated	Not specified	16 / Not applicable	Control board and extensive air devices
Hu <i>et al.</i> [8]	2023	Linkage and bevel gear	Not specified	13 / Not specified	Wrist-required
Cheng <i>et al.</i> [9]	2023	Linkage and gear	180×85	10 / 15	Wrist-required
ManoPla hand [20]	2023	Linkage, rack and pinion	Not specified	17 / Not specified	All integrated
Blue Hand [22]	2023	Pneumatic actuated	$218 \times 123 \times 32$	21 / Not applicable	Control board and extensive air devices
The proposed hand	2025	Linkage and pneumatic actuated	$195 \times 112.1 \times (24-40)$	10 / 16	Control board and air devices

* First introduced in 2005, with ongoing upgrades.

A significant challenge in evaluating and comparing robotic hand designs lies in the diversity of their design principles, performance metrics, and assessment criteria, which makes establishing a fair and unified benchmarking framework difficult.

From the perspective of actuation, pneumatic hands demonstrate a considerable advantage over rigid ones in grasping irregular or fragile objects, as they can adapt without a sophisticated control system. However, their inherent reliance on external components like pumps and valves limits their overall practicality and portability. When the entire actuation system is considered, a pneumatic hand holds no distinct advantage over a linkage-driven, or tendon-driven hand. Furthermore, implementing precise force control with pneumatic actuation remains challenging.

The lack of standardized performance evaluation presents a major obstacle in comparing robotic hand designs. Key metric such as workspace volume is not generally provided across different studies. While a design with longer finger phalanges may yield a larger workspace, this feature alone does not necessarily indicate a superior design. Moreover, established assessment methods like the Kapandji test or the Feix taxonomy are

not universally adopted. Even when evaluated under the same standard, the assessment whether a specific hand posture is fully achievable or merely visually similar depends heavily on the authors' subjective judgment. In cases where certain degrees of freedom are absent, some gestures are theoretically impossible to fully realize, as discussed in our analysis.

Cost is another critical dimension for comparison. Rigid hands, typically driven by linkages or gears, costs highly due to the precision machining of metal components. In contrast, the casting molds of a soft hand is often substantially lower. However, this advantage may be offset by the cost of air pumps and valves. Generally speaking, rigid hands are often more compact but costly, while soft hands offer a lower cost at the expense of a larger overall volume.

5 Supplementary videos

Concerning the repeatability of the test grasps, we have provided [supplementary videos](#) that demonstrate the maneuverability and versatility of our hand. These visual materials, when considered together with the experimental figures in Fig. 8, facilitate a straightforward assessment of the effectiveness towards varied grasps.

[Supplementary Video.mp4](#) showcases the motor-driven degrees of freedom. [Supplementary Video 2.mp4](#) showcases the pneumatic actuation.

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